



DATA ANALYST TAKE HOME TEST

Surge Pricing Analysis

By Fayzan Rizky Hidayat

Understanding the factors behind dynamic pricing in taxi aggregator platform.



Dataset

131,662 Trips



Analysis

8 Key Questions

April 2026

Introduction



Fayzan Rizky Hidayat

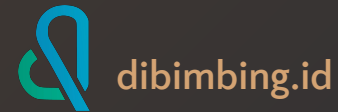
Data Science & Data Analyst Enthusiast



Self-Overview

Physics graduate transitioning into Data Science, with strong analytical and problem-solving skills. Experienced in Python, statistics, and data visualization. Currently joining Dibimbing.id Data Analyst & Data Science Bootcamp to enhance technical expertise and apply data-driven insights in real-world projects. Eager to contribute to data analysis and business decision-making processes.

Education Background



Data Analyst / Data Science Bootcamp
September 2025 - 2026



Universitas Padjadjaran

Bachelor of Physics Sciences
Agustus 2021 - Agustus 2025

Executive Overview

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Project Background

Business context, problem statement, and project objectives

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Dataset Overview

Data characteristics, preprocessing, and feature dictionary

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Univariate Analysis

Feature distributions and initial data exploration

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Bivariate Analysis

8 analytical questions examining factor relationships

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Multivariate Analysis

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Key Findings

Summary of significant and non-significant factors



Strategic Recommendations

Actionable business strategies for pricing optimization, operational excellence, and customer experience enhancement

Final Section

07

Project Background & Business Context



Company Description

Sigma Cabs is a leading **taxi aggregator platform in India** that connects users with various taxi service providers. As an intermediary, the platform matches customer needs with the best available ride options from multiple partner providers.



The Challenge

The company faces significant challenges with **dynamic surge pricing** determined by service providers. After nearly a year of operations, Sigma Cabs struggles to identify the dominant factors triggering surge pricing increases (scale 1-3).



Project Objectives

Generate **data-driven insights** to identify key factors influencing surge pricing, enabling improved pricing strategies, supply-demand optimization, and enhanced customer experience.



Business Problem

Inability to understand surge pricing factors leads to:

- ✗ **Inefficiency** in matching customers with appropriate services
- ✗ **Decreased customer satisfaction** due to unpredictable pricing
- ✗ **Operational inefficiencies** affecting company performance

“ Problem Statement

"What are the main factors that influence Surge Pricing Type, and how can trip patterns and customer characteristics be used to understand and anticipate price increases?"

Dataset Overview & Preprocessing

Dataset Characteristics

131,662

Total Records

44.2 km

Average Distance

36.2%

High Surge (Type 3)

2.85 / 5

Average Rating

Feature	Type	Description
Trip_Distance	float64	Distance requested by customer (km)
Type_of_Cab	object	Cab category requested (A-E)
Customer_Since_Months	float64	Customer tenure in months
Life_Style_Index	float64	Proprietary lifestyle index
Confidence_Life_Style_Index	object	Category showing confidence on the lifestyle index
Destination_Type	object	Destination category (A-N, 14 types)
Customer_Rating	float64	Lifetime average rating
Cancellation_Last_1Month	int64	Trips cancelled in last month
Var1, Var2, Var3	float64	Continuous variables masked by the company.
Gender	object	Gender of the customer
Surge_Pricing_Type	int64	Target: Surge type (1, 2, or 3)

Preprocessing Steps

- Dropped Features

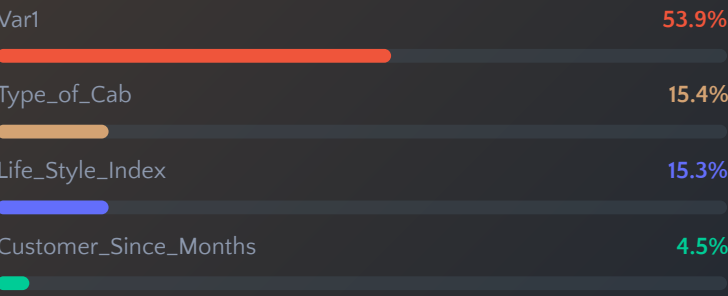
Trip_ID (identifier) and Var1 (54% missing)
- Categorical Imputation

Mode for Type_of_Cab & Confidence_Life_Style_Index
- Numerical Imputation

Median for Customer_Since_Months & Life_Style_Index
- Type Conversion

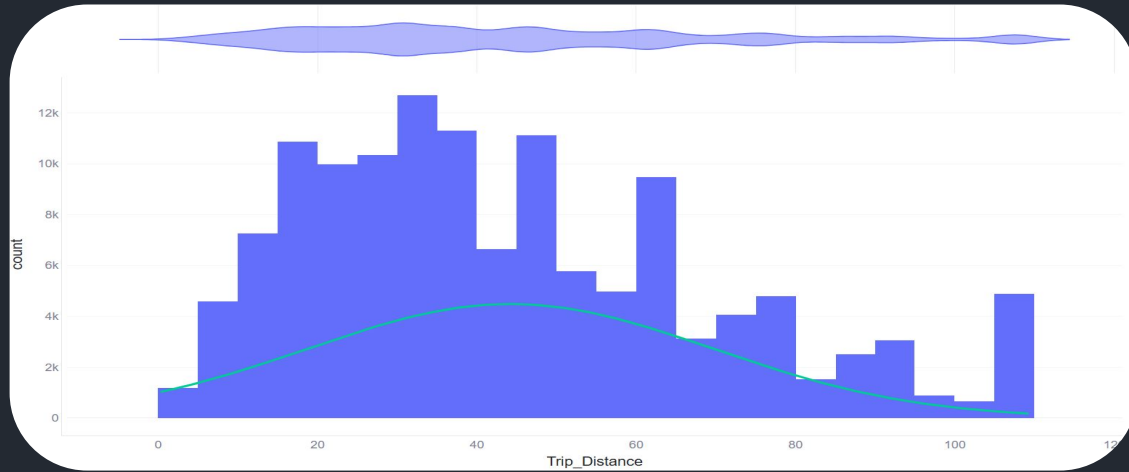
Surge_Pricing_Type → Categorical

Missing Value Summary

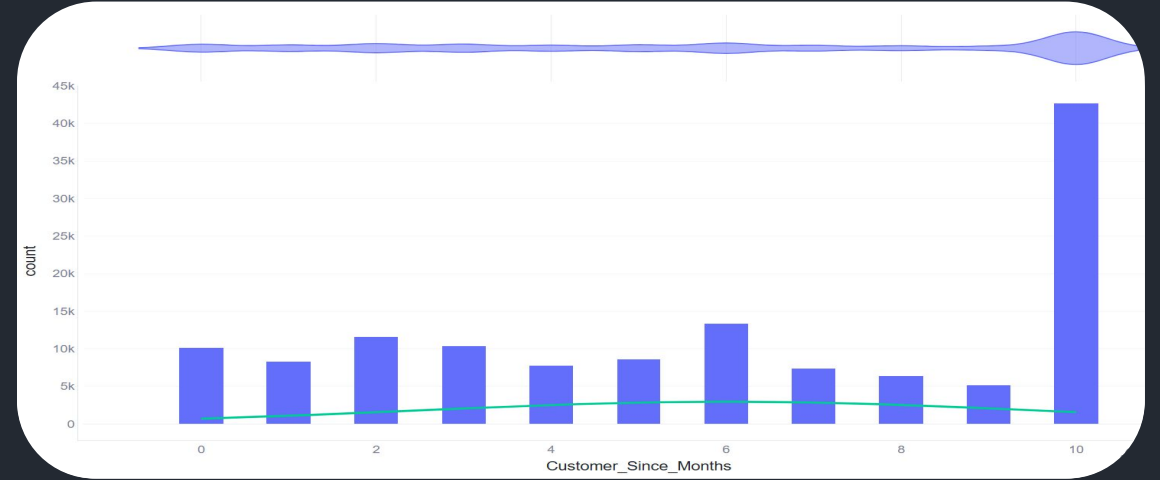


Numerical Feature Distributions

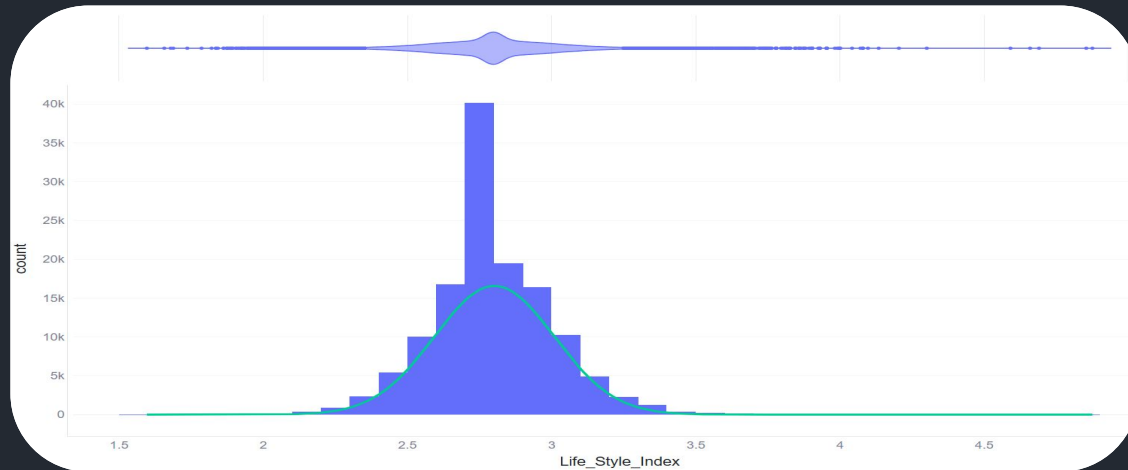
Trip_Distance



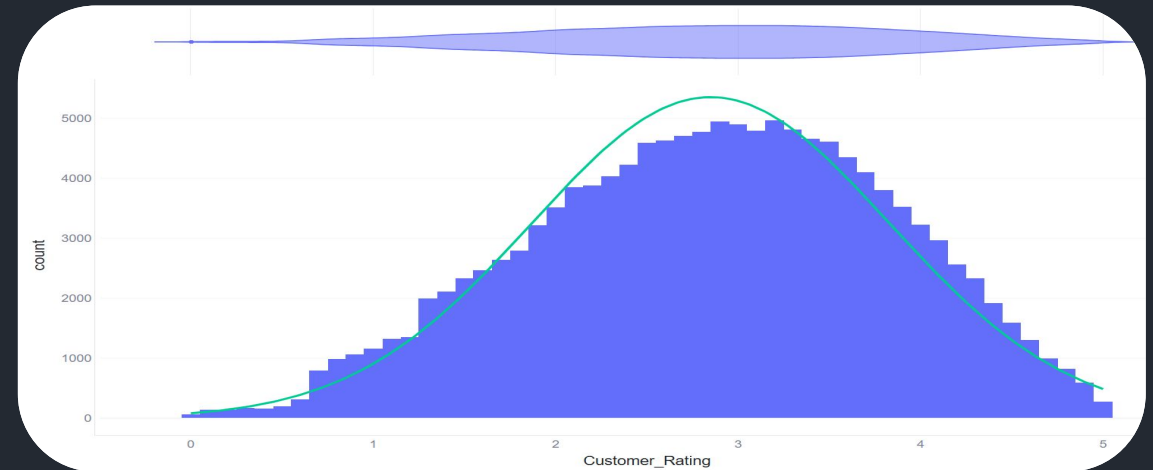
Customer_Since_Months



Life_Style_Index

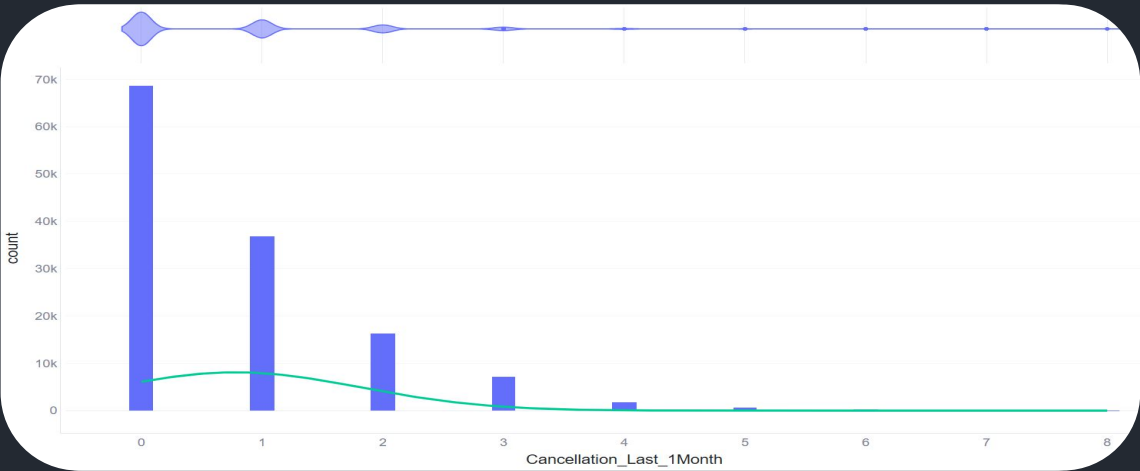


Customer_Rating

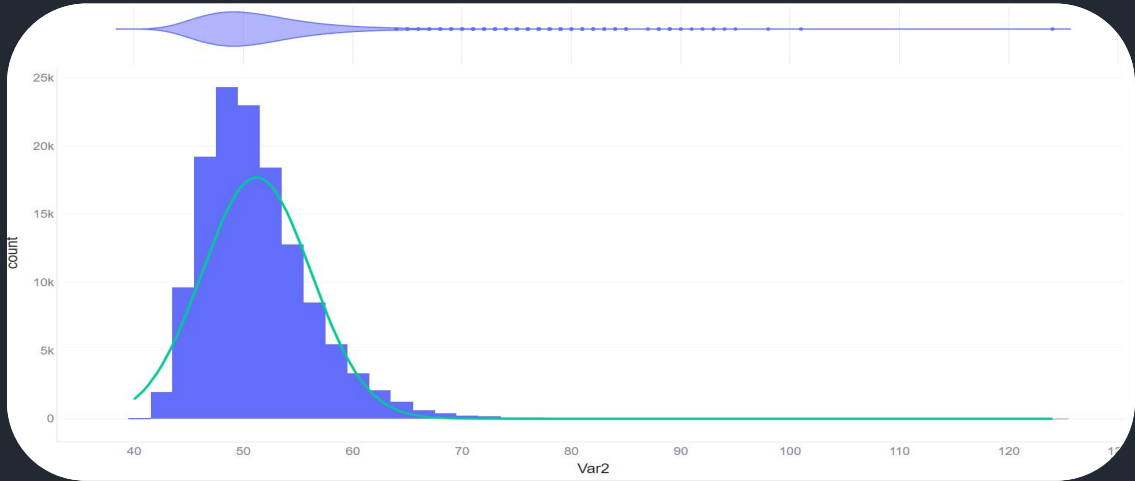


Numerical Feature Distributions

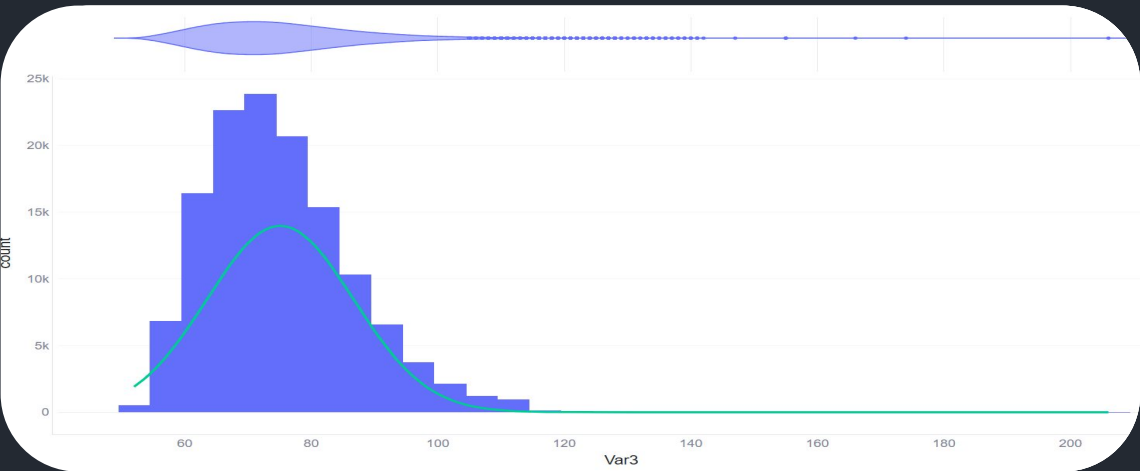
Cancellations_Last_1Month



Var2

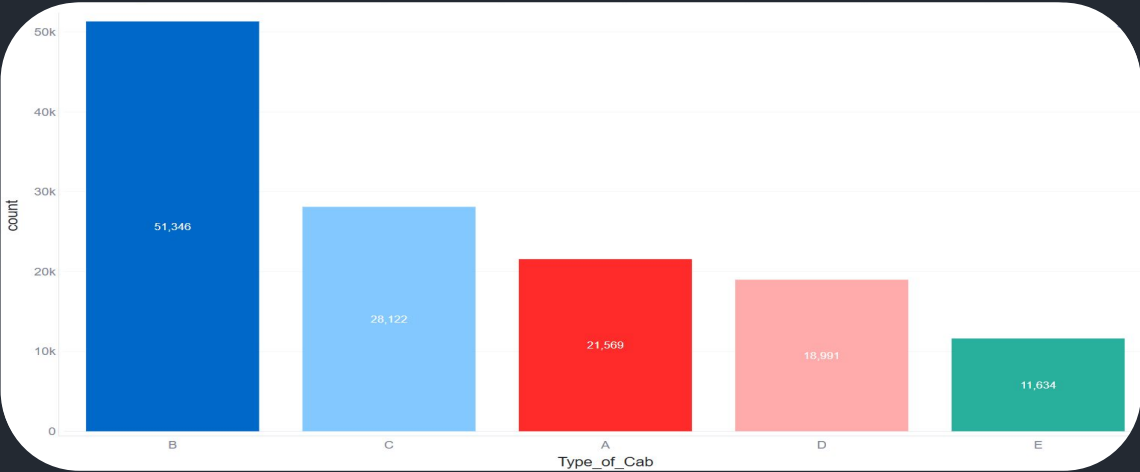


Var3

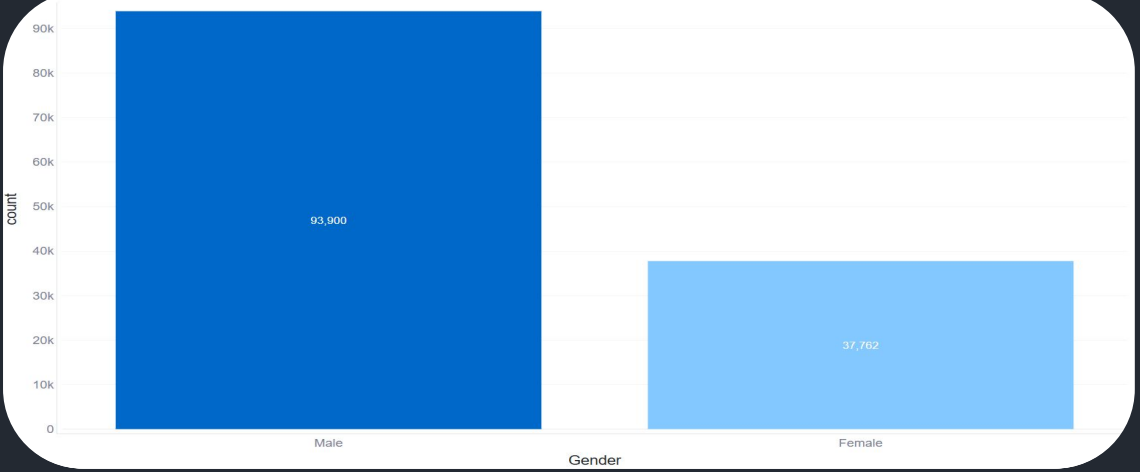


Categorical Feature Distributions

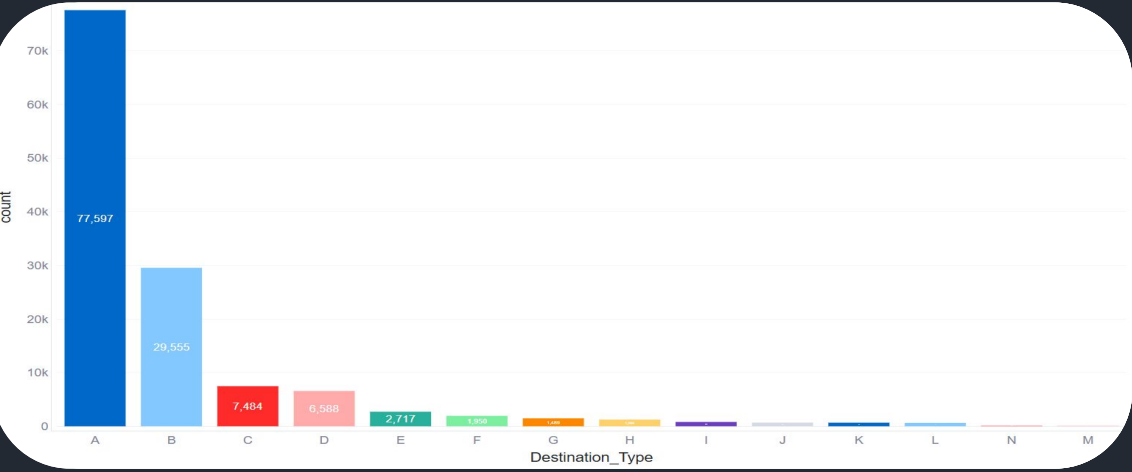
Type_of_Cab



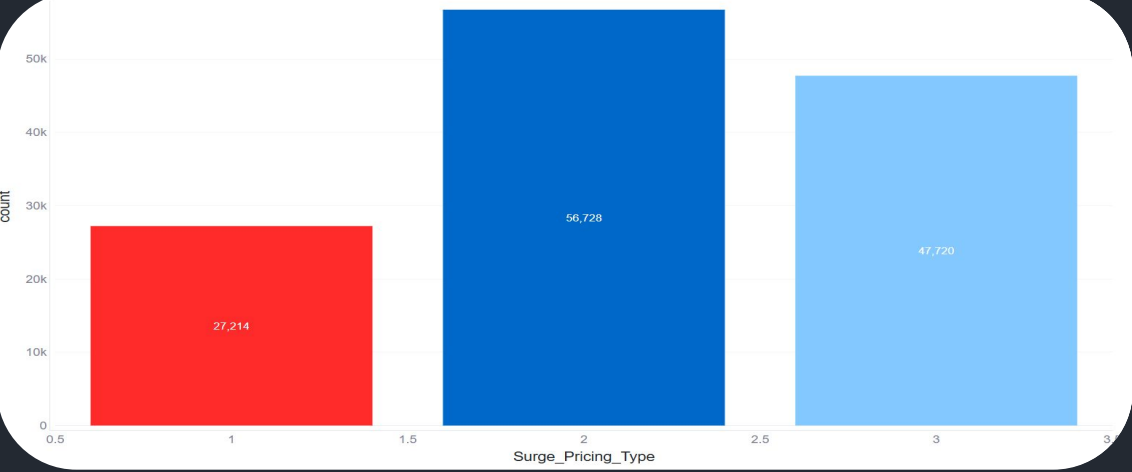
Gender



Destination_Type



Surge_Pricing_Type



Key Findings

Numerical Features

Trip_Distance

Normal

Mean: 38.2 km

Median: 36.0 km

Customer_Rating

Normal

Mean: 2.85

Median: 2.90

Life_Style_Index

Normal

Mean: 2.80

Median: 2.98

Var2 & Var3

Skewed

Require transformation

Numerical Features Key Observations

- ✓ Var2 and Var3 show skewed distributions requiring transformation for modeling
- ✓ Trip distances show multimodal patterns indicating popular route clusters
- ✓ Customer ratings are normally distributed around 2.9/5.0

Categorical Features

Surge_Pricing_Type

20.7%

Type 1

43.1%

Type 2

36.2%

Type 3

Type 2 is the most frequent category

Type_of_Cab

Type A: 21.4%

Type B: 39%

Type C: 16.4%

Type D: 14.4%

Type E: 8.8%

Gender

Male

71.3%

Female

28.7%

Destination_Type

Type A: 58.9%

Type B: 22.4%

Type C: 5.7%

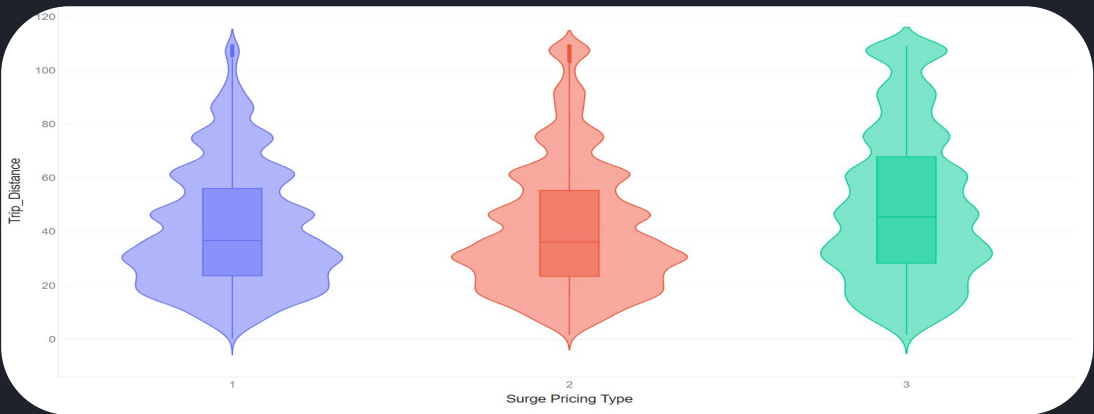
Others (Type D-N): 13%

Categorical Features Insights

The dataset is heavily dominated by male customers, who account for 71.3% of total users. Type B cabs are the most utilized, capturing a 39% market share. Destination Type A represents an overwhelming 58.9% of all trips. Furthermore, Surge Type 2 is the most frequent pricing condition, while Type 1 remains the rarest.

Trip Distance & Cab Type Impact

Q1 Does trip distance affect surge pricing?



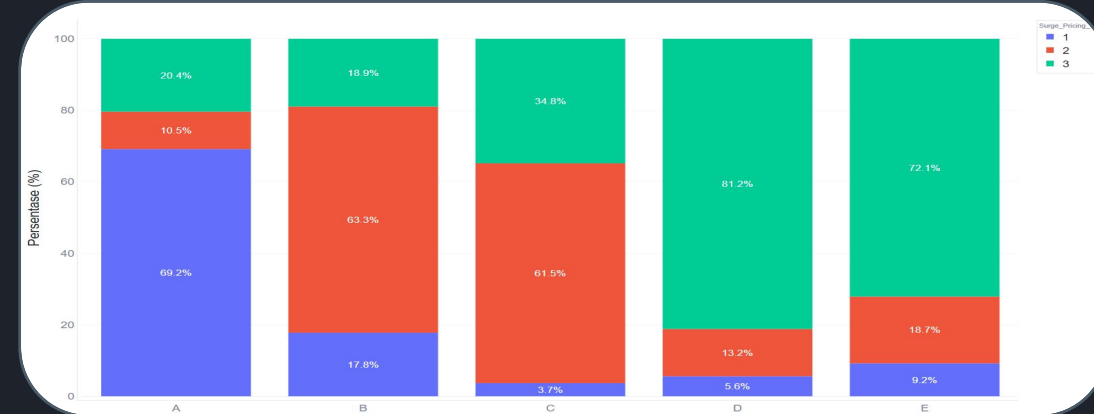
Type 1: Median 36 km
Standard pricing range

Type 2: Median 36 km
Similar to Type 1

Type 3: Median 45 km
Significantly higher (+25%)

Insight: Longer trips have higher probability of Type 3 surge pricing, with multimodal distribution showing popular route clusters at 30, 45, 60, and 75 km.

Q2 Does cab type influence surge pricing?



Type A (Standard) 69% Type 1
Most stable pricing

Type B & C (Mid-range) 61-63% Type 2
Moderate surge dominant

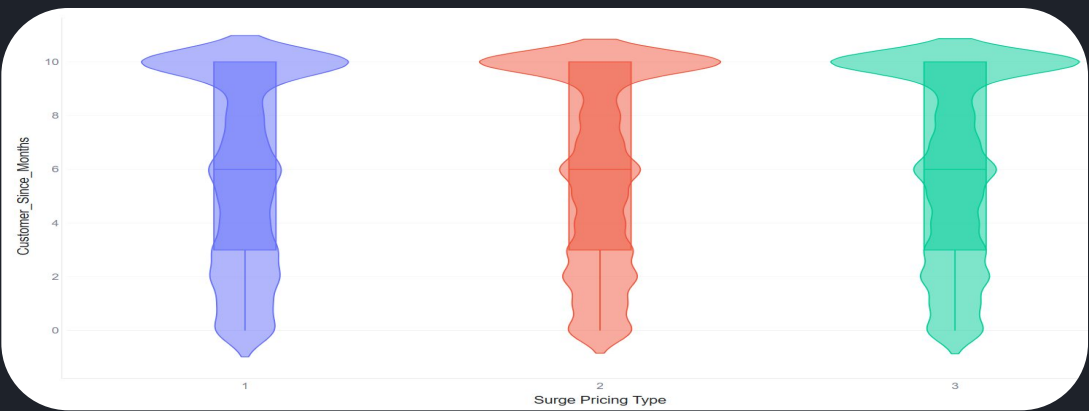
Type D & E (Premium) 72-81% Type 3
Highest surge probability

Key Finding: Type_of_Cab is the strongest determinant with clear linear pattern – higher cab class = higher surge probability.

Customer Loyalty & Lifestyle Analysis

Q3

Do new and old customers have different surge patterns?



Statistical Profile

Median

6

Q1

2

Q3

10

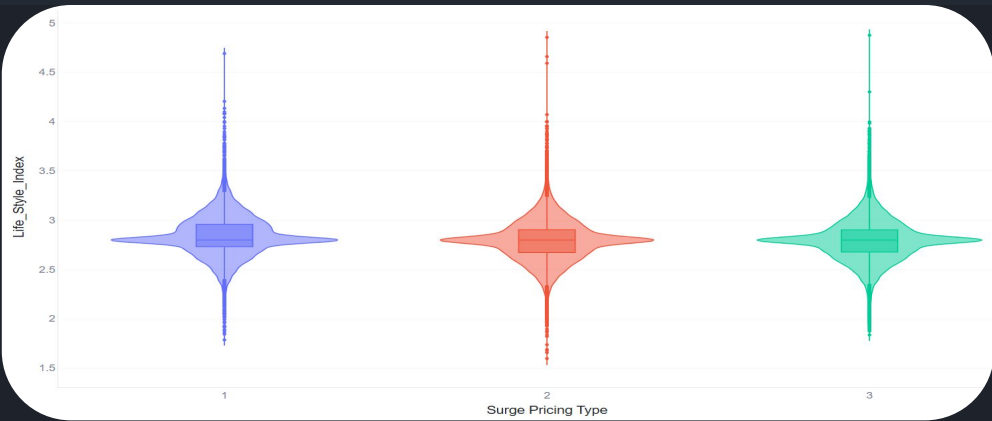
IDENTICAL

Same across all surge types

✗ **Conclusion:** Customer loyalty (subscription duration) is NOT a determinant of surge pricing. No difference between new and existing customers.

Q4

How does customer lifestyle affect pricing?



Statistical Profile

Median

2.98

Mean

2.80

Mode

2.98

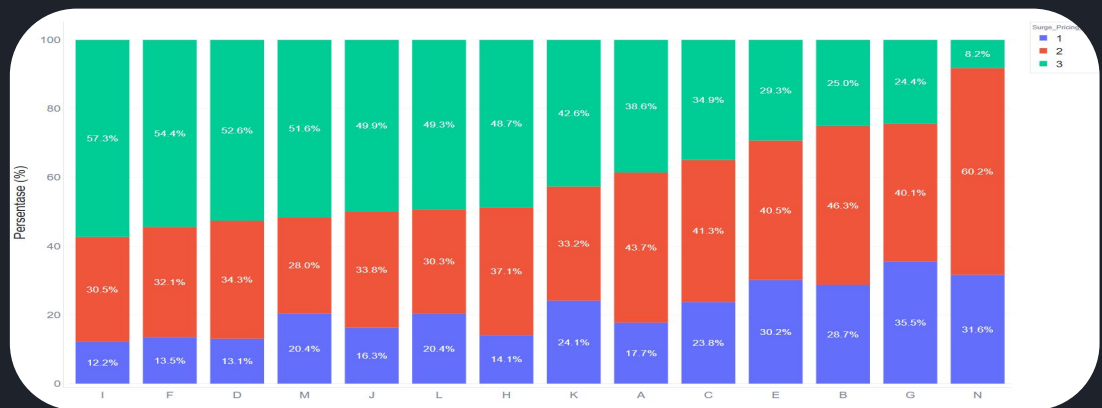
IDENTICAL

Same across all surge types

✗ **Conclusion:** Life_Style_Index is NOT a factor in surge pricing determination. All customer lifestyle profiles show identical patterns.

Destination Type & Customer Rating Impact

Q5 Does destination type affect surge pricing?



High Surge Cluster

Majority destinations dominated by Type 3 surge

Moderate Surge Cluster

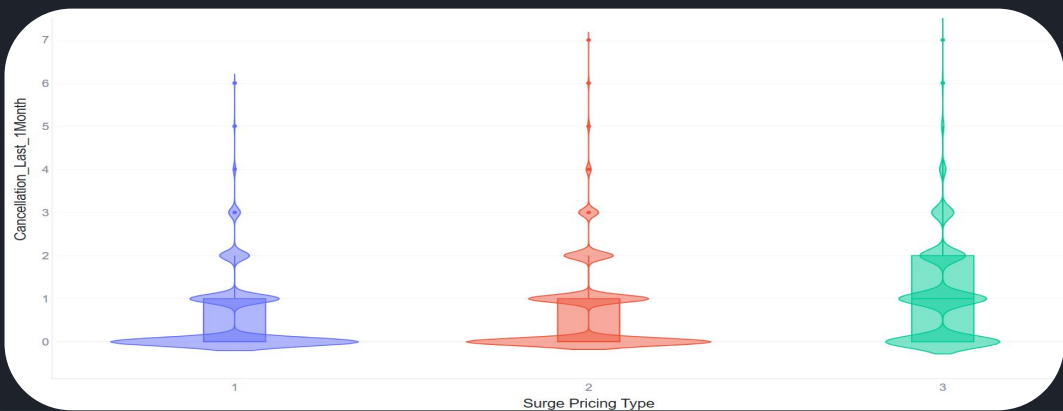
Destinations A, B, C, E, G, N dominated by Type 2

Critical Finding

No destination shows Type 1 dominance

Insight: Normal pricing (Type 1) is extremely rare across all destinations, indicating consistent demand pressure or supply constraints.

Q6 How does customer rating affect pricing?



Type 1: Median 3.05
Highest ratings

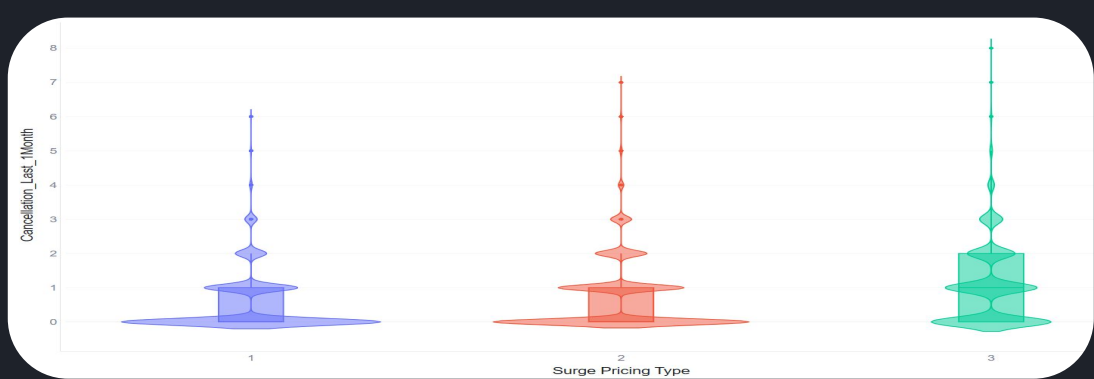
Type 2: Median 2.90
Moderate ratings

Type 3: Median 2.75
Lowest ratings

Key Finding: Clear negative correlation – higher customer ratings associate with lower surge types. Rating is a significant factor.

Cancellation History & Gender Analysis

Q7 How does cancellation behavior affect pricing?



Type 1: Median 0
Majority have no cancellations

Type 2: Median 0
Similar to Type 1

Type 3: Median 1
Higher cancellation frequency

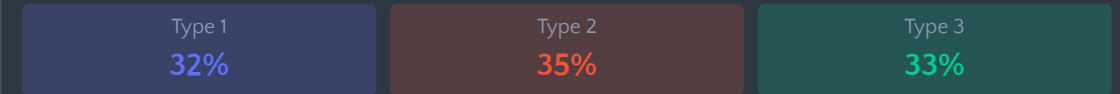
⚠ Extreme Cases (≥3)
Type 3 dominates cancellations of 3 or more

📌 Key Finding: Positive correlation – higher cancellation frequency associates with higher surge types. Cancellation history is a significant factor.

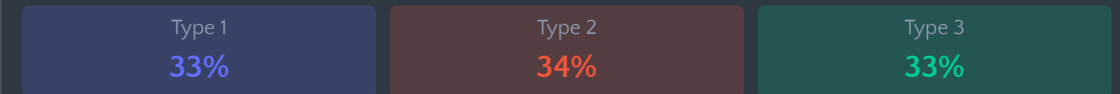
Q8 Are surge pricing proportions equal across genders?



Female Customers 28.7%



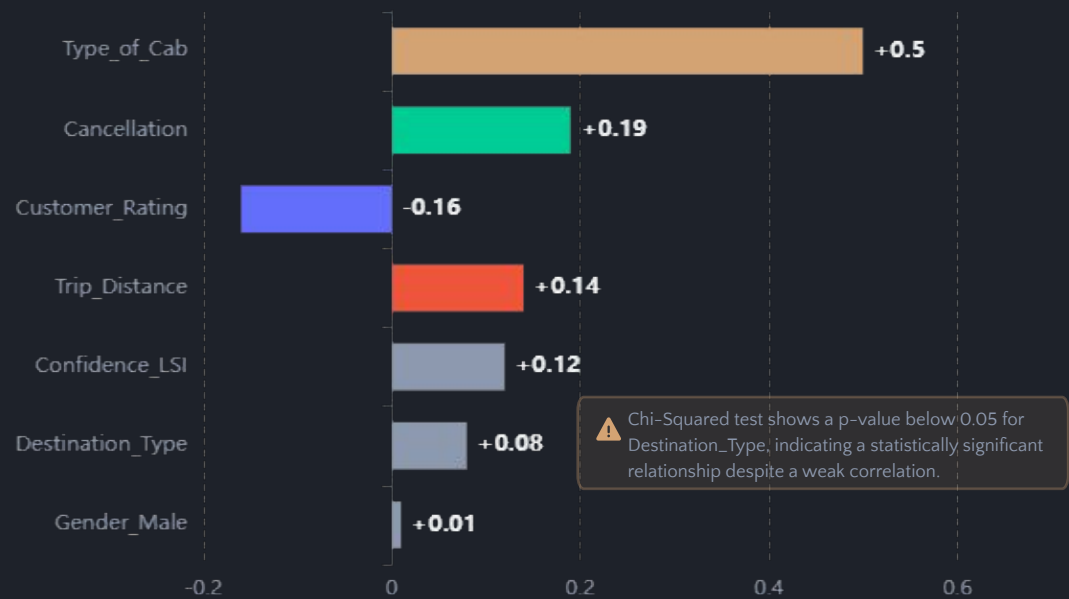
Male Customers 71.3%



✖ Conclusion: Gender shows NO significant difference. Surge pricing proportions are nearly identical between female and male customers.

Correlation Insights & Feature Interactions

Feature Correlation with Surge Pricing



Strongest Predictors

Type_of_Cab 0.50
Strongest determinant

Cancellation_Last_1Month 0.19
Positive correlation

Customer_Rating -0.16
Negative correlation

Trip_Distance 0.14
Moderate positive

Feature-to-Feature Insights

- Var2 ↔ Var3: 0.68
- Var3 ↔ Life_Style_Index: 0.30
- Confidence_Life_Style_Index ↔ Trip_Distance: 0.22
- Trip_Distance ↔ Life_Style_Index: 0.47
- Var3 ↔ Customer_Rating: -0.23
- Trip_Distance ↔ Destination_Type: -0.17
- Var2 ↔ Customer_Rating: -0.30
- Trip_Distance ↔ Var3: 0.23
- Life_Style_Index ↔ Customer_Rating: 0.23

Key Findings Summary

8 Analytical Questions - Impact Assessment

Question	Factor	Impact	Key Insight
Q1	Trip Distance	SIGNIFICANT	Type 3 median 45km vs 36km for Types 1&2
Q2	Type of Cab	STRONGEST	Linear pattern: A→Type 1, D/E→Type 3 (81%/72%)
Q3	Customer Since Months	NOT SIGNIFICANT	Identical medians (6 months) across all types
Q4	Life Style Index	NOT SIGNIFICANT	Identical median (2.98) across all types
Q5	Destination Type	SIGNIFICANT	While correlation is low, p-value < 0.05
Q6	Customer Rating	SIGNIFICANT	Negative correlation: higher rating → lower surge
Q7	Cancellation History	SIGNIFICANT	≥3 cancellations dominated by Type 3
Q8	Gender	NOT SIGNIFICANT	Identical proportions across Female/Male



Significant
Factors

5

factors significantly influence surge pricing



Non-Significant

3

factors show no meaningful impact

Correlation Ranking

Type_of_Cab

0.50

Cancellation

0.19

Customer_Rating

-0.16

Trip_Distance

0.14

Destination_Type

0.08

Dominant Factor

Type_of_Cab emerges as the strongest predictor with **r=0.50**, indicating that cab classification alone explains significant variance in surge pricing.

Business Strategy & Implementation



Pillar 1

Pricing Strategy

🎯 Cab-Based Segmentation

Use Type_of_Cab as primary segmentation factor. Implement transparent surge communication per vehicle class.

📈 Premium Customer Transparency

Build trust with premium customers (Type D/E) through clear surge explanations and loyalty benefits.

⚖️ Dynamic Distance Pricing

Factor trip distance into surge algorithms, with higher thresholds for longer journeys.



Pillar 2

Operational Excellence

⚠️ Cancellation Monitoring

Implement early warning systems for high-cancellation customers. Flag accounts with many cancellations for special handling.

🔄 Supply-Demand Optimization

Use cancellation patterns to predict demand spikes and pre-position drivers in high-risk areas.

📍 Destination Intelligence

Focus operational resources on "Moderate Surge" destinations (A, B, C, E) which drive highest transaction volumes.



Pillar 3

Customer Experience

★ Rating-Based Loyalty Programs

Reward high-rated customers with surge discounts. Implement tiered benefits based on lifetime rating performance.

🎁 Behavioral Incentives

Encourage positive behavior through rewards: low cancellation rates, high ratings, and consistent usage.

♥️ Retention Focus

Since loyalty duration doesn't affect pricing, focus on quality of service rather than tenure-based benefits.



Implementation Priority

Start with **Cab-Based Segmentation** (highest impact, $r=0.50$) and **Cancellation Monitoring** (immediate operational benefit). These two initiatives alone address the strongest predictive factors identified in the analysis. Roll out **Rating-Based Programs** as a medium-term initiative to improve customer satisfaction and retention.

Key Takeaways & Next Steps

Core Findings

- 5

Key Factors Identified

Type_of_Cab, Cancellation, Customer_Rating, Trip_Distance, Destination_Type influence surge
- 

Dominant Predictor

Type_of_Cab with $r=0.50$ is the strongest determinant of surge pricing type
- 3

Non-Significant Factors

Customer_Since_Months, Life_Style_Index, Gender show no meaningful impact

Data-Driven Foundation

This analysis provides a **robust foundation** for building transparent, fair, and data-driven pricing strategies. By focusing on the 4 significant factors, Sigma Cabs can optimize both revenue and customer satisfaction.

Recommended Next Steps

1. Predictive Modeling

Develop machine learning models to anticipate surge pricing before trip requests
2. Real-Time Optimization

Implement dynamic pricing algorithms that respond to real-time supply-demand
3. A/B Testing

Test pricing transparency initiatives and loyalty program effectiveness
4. Driver Incentives

Align driver incentives with surge patterns to optimize fleet distribution

Expected Impact

 Improved pricing accuracy, enhanced customer trust, and optimized operational efficiency